

Printing student marks memo:

```
#include<stdio.h>

#include<conio.h>

void main()

{

int id, sub[6], tot=0, p=1, i;

char name[30], father[30],mother[30],dob[20],gender[10],school[40],

id1[30], id2[30], subj[6][10]={"Tel","Eng","Hin","Mat","Sci","Soc"};

float avg;

clrscr();

printf("Enter stu id "); scanf("%d",&id);flushall();

printf("Enter stu name "); gets(name);

printf("Enter father name "); gets(father);

printf("Enter mother name "); gets(mother);

printf("Enter gender "); gets(gender);

printf("Enter date of birth "); gets(dob);

printf("Enter school name "); gets(school);

printf("Enter 1st identification mark "); gets(id1);

printf("Enter 2nd identification mark "); gets(id2);
```

```
printf("Enter 6 sub marks ");
for(i=0;i<6;i++)
{scanf("%d",&sub[i]); if(sub[i]<35)p=0; tot+=sub[i];}
avg=tot/6.0;
puts("-----");
puts("\t\t BOARD OF SECONDARY EDUCATION");
puts("\t\t\t ANDHRA PRADESH");
puts("-----");
printf("Rollno: %d\t Name: %s\n",id,name);
printf("Father: %s\t Mother: %s\n",father,mother);
printf("DOB: %s\t Gender: %s\n",dob,gender);
printf("School: %s\n",school);
puts("-----");
puts("Identification Marks");
puts("-----");
printf("1. %s\n",id1);
printf("2. %s\n",id2);
puts("-----");
puts("Subj\tMarks\tPass/Fail");
puts("-----");
```

```
for(i=0;i<6;i++)  
{  
printf("%s\t%d\t%s\n",subj[i], sub[i], sub[i]>=35?"P":"F");  
}  
puts("-----");  
printf("%s tot=%d, avg=%.2f and  
got %s",name,tot,avg,p?"Passed":"Failed");  
getch();  
}
```

```
TC
Enter stu id 1
Enter stu name krish
Enter father name abc
Enter mother name xyz
Enter gender male
Enter date of birth 1-1-2000
Enter school name zpph school
Enter 1st identification mark a mole on right shoulder
Enter 2nd identification mark a mole on fore head
Enter 6 sub marks 88 99 80 98 98 88
-----
BOARD OF SECONDARY EDUCATION
ANDHRA PRADESH
-----
Rollno: 1      Name: krish
Father: abc    Mother: xyz
DOB: 1-1-2000  Gender: male
School: zpph school
-----
Identification Marks
-----
1. a mole on right shoulder
2. a mole on fore head
-----
2. a mole on fore head
-----
subj      Marks      Pass/Fail
-----
Tel       88         P
Eng       99         P
Hin       80         P
Mat       98         P
Sci       98         P
Soc       88         P
-----
krish tot=551, avg=91.83 and got Passed_
-----
```

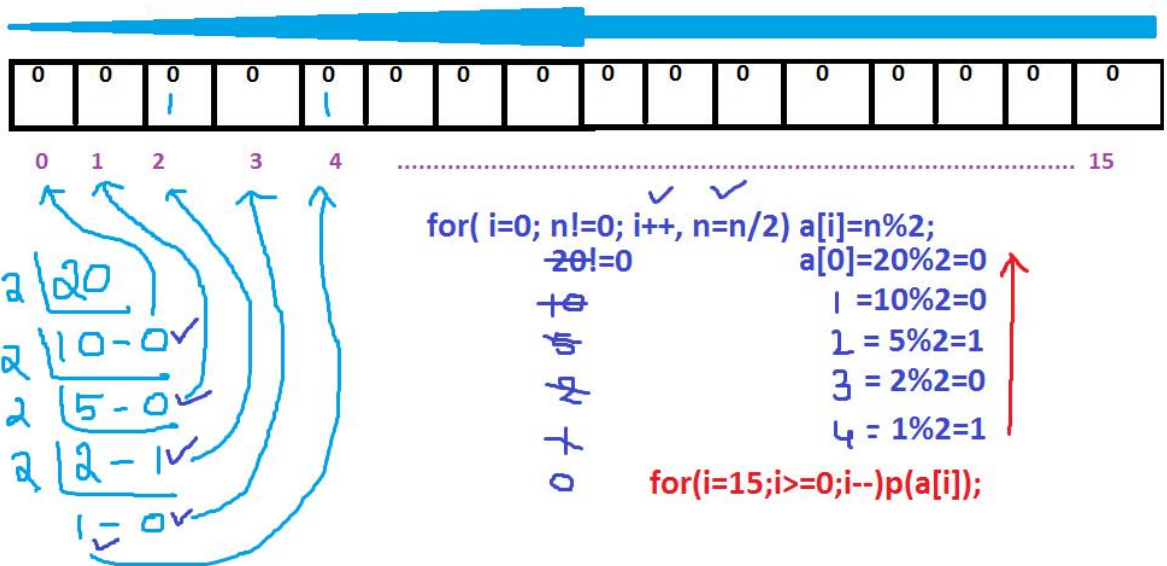
Decimal to binary conversion:

Eg: 20 → 0000 0000 0001 0100

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 61 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[16]={0},n,i;
clrscr();
printf("Enter a no ");scanf("%d",&n);
for(i=0;n!=0;i++,n=n/2)a[i]=n%2;
printf("Binary code ");for(i=15;i>=0;i--)printf("%2d",a[i]);
getch();
}

Enter a no 20
Binary code  0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0 0_

TC
2:29 PM
20-Sep-24
```



Decimal to octal conversion using a user defined program:

$$8 \overline{) 20} \\ 2-4 \checkmark$$

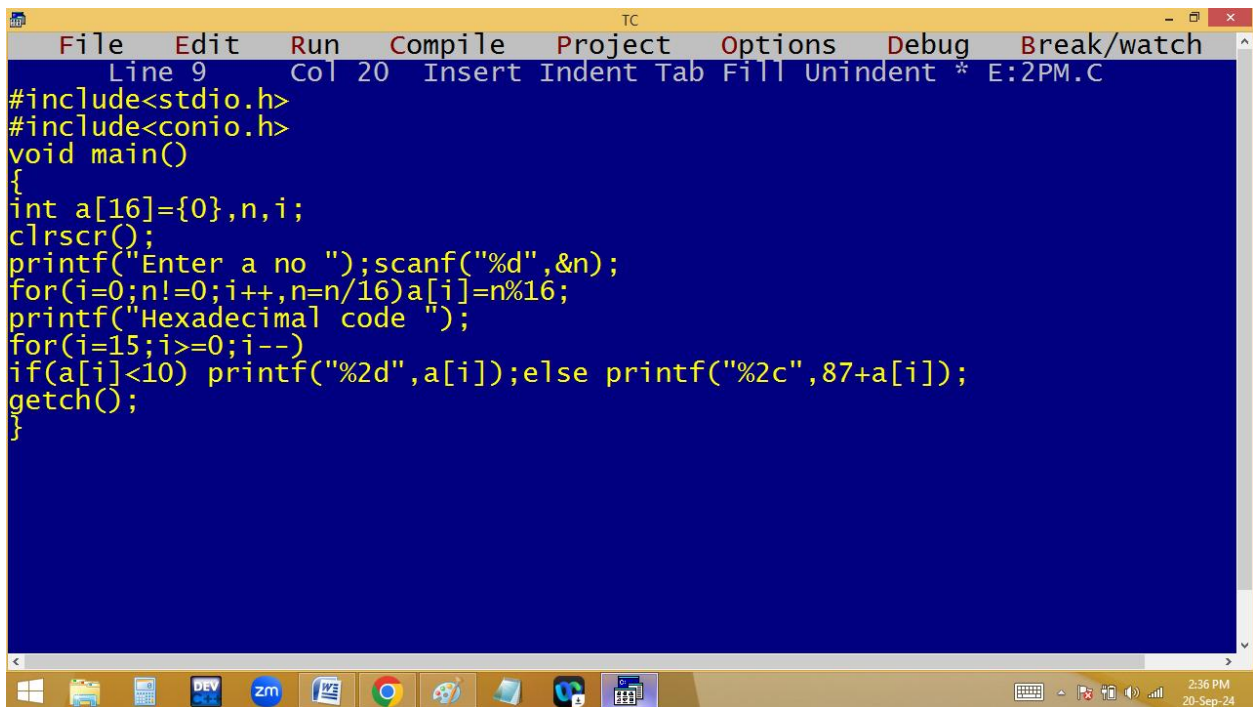
The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code for a program that converts a decimal number to its octal representation. The code uses an array to store the octal digits and prints them in reverse order. The bottom screenshot shows the program's execution: the user enters the decimal number 20, and the program outputs the octal code 0000000000000024.

```
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 32 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[16]={0},n,i;
clrscr();
printf("Enter a no ");scanf("%d",&n);
for(i=0;n!=0;i++,n=n/8)a[i]=n%8;
printf("Octal code ");for(i=15;i>=0;i--)printf("%2d",a[i]);
getch();
}
```

Enter a no 20
Octal code 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 4

Decimal to hexadecimal conversion:

$$\begin{array}{r} 16 \overline{) 20} \\ 1-4 \checkmark \end{array}$$



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 20 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[16]={0},n,i;
clrscr();
printf("Enter a no ");scanf("%d",&n);
for(i=0;n!=0;i++,n=n/16)a[i]=n%16;
printf("Hexadecimal code ");
for(i=15;i>=0;i--)
if(a[i]<10) printf("%2d",a[i]);else printf("%2c",87+a[i]);
getch();
}
```

```
TC
Enter a no 45
Hexadecimal code 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 d
```

```
TC
Enter a no 95
Hexadecimal code 0 0 0 0 0 0 0 0 0 0 0 0 0 0 5 f
```

```
TC
Enter a no 20
Hexadecimal code 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 4_
```

0 9 10-a 11-b 12-c 13-d 14-e 15-f

Arranging array elements in reverse order:

Temporary:

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 61 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i;
clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n); for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Reverse order");for(n--;n>=0;n--)printf("%3d",a[n]);_
getch();
}

Enter array size 1-100 7
Enter 7 elements 2 0 3 7 1 5 6
Reverse order 6 5 1 7 3 0 2

TC
2:42 PM
20-Sep-24
```

```
for( ✓n--; n >= 0; ✓n-- ) p(a[n]);
```

1 0 2 3 5

a

9	0	7	5
0	1	2	3

5 7 0 9 ✓

Permanent method:

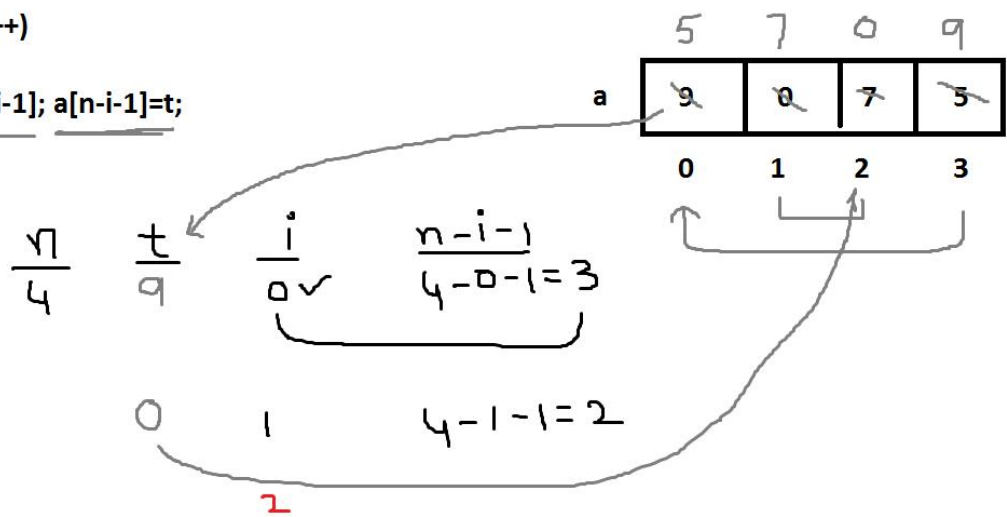
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 33 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i,t;
clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n); for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Reverse order");
for(i=0;i<n/2;i++){t=a[i];a[i]=a[n-i-1];a[n-i-1]=t;}
for(i=0;i<n;i++)printf("%3d",a[i]);
getch();
}

Enter array size 1-100 4
Enter 4 elements 1 0 2 4
Reverse order 4 2 0 1_

TC
2:54 PM
20-Sep-24
```

```
Enter array size 1-100 5
Enter 5 elements 1 0 2 8 5
Reverse order 5 8 2 0 1_
```

```
for( i=0; i<n/2; i++)
{
    t=a[i]; a[i]=a[n-i-1]; a[n-i-1]=t;
}
```



Without using 3rd variable:

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 62 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i,t;
clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n); for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Reverse order");
for(i=0;i<n/2;i++)
{a[i]=a[i]+a[n-i-1];a[n-i-1]=a[i]-a[n-i-1];a[i]=a[i]-a[n-i-1];}
for(i=0;i<n;i++)printf("%3d",a[i]);
getch();
}
```

Enter array size 1-100 5
Enter 5 elements 1 2 3 4 5
Reverse order 5 4 3 2 1

```
for( i=0;i<n/2;i++)
{
```

```
    a[i]=a[i]+a[n-i-1];
```

```
    a[n-i-1]=a[i]-a[n-i-1];
```

```
    a[i]=a[i]-a[n-i-1];
```

```
}
```

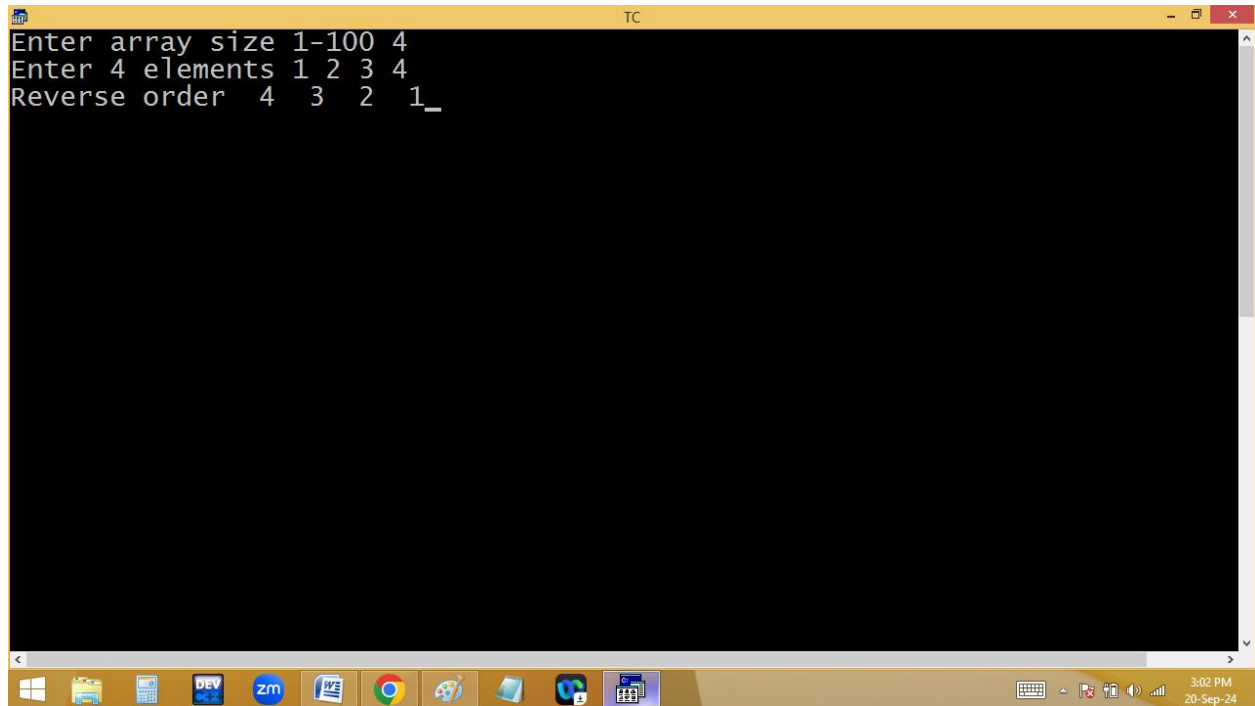
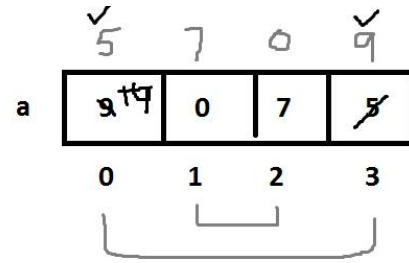
$$i = i + \frac{n-i-1}{4} \quad \frac{i}{0}$$

$$0 + 3 = 9 + 5 = 14$$

$$n-i-1 = \frac{i}{3} - n-i-1$$

$$3 \quad 14 - 5 = 9$$

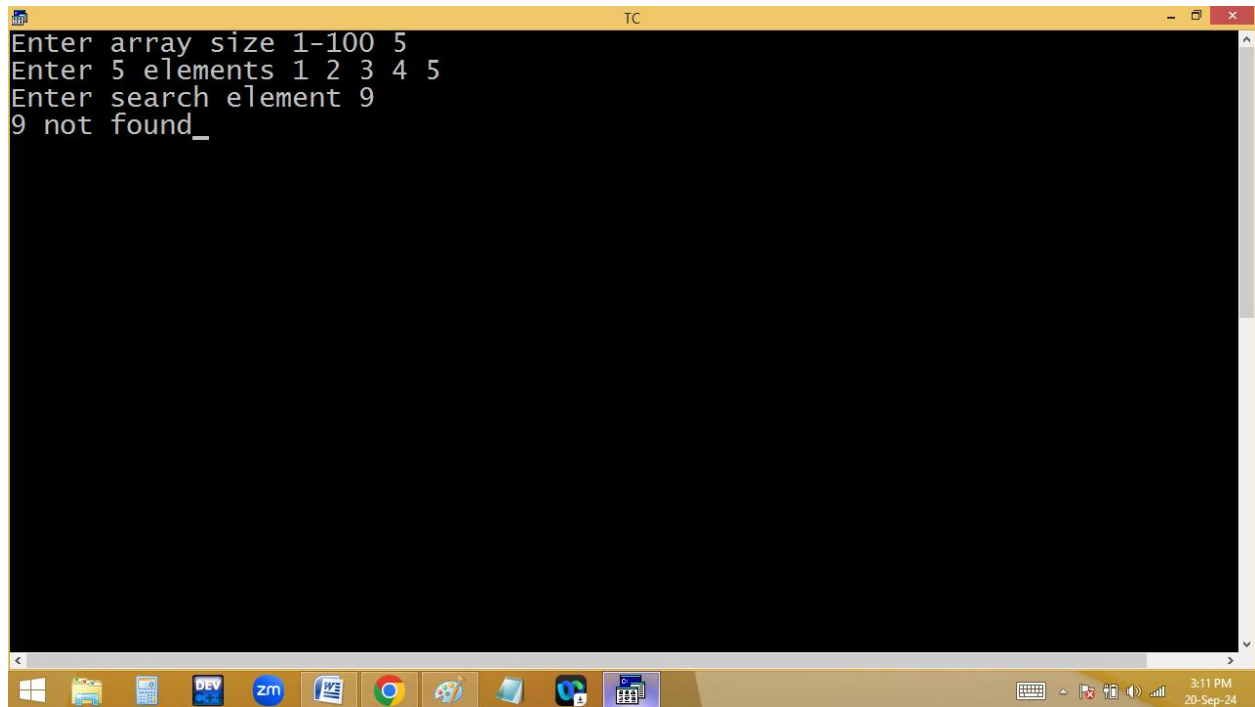
$$i = i - n-i-1 = 14 - 9 = 5$$



Linear search:

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 14 Col 30 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i,ele,f=0;
clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n); for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Enter search element "); scanf("%d",&ele);
for(i=0;i<n;i++)
{
if(a[i]==ele) printf("%d in %d cell\n",ele,i+1,f=1);
}
if(f==0)printf("%d not found",ele);
getch();
}
```

Enter array size 1-100 9
Enter 9 elements 1 0 2 1 7 5 1 9 4
Enter search element 1
1 in 1 cell
1 in 4 cell
1 in 7 cell



```
TC
Enter array size 1-100 5
Enter 5 elements 1 2 3 4 5
Enter search element 9
9 not found_
```

Printing Cell no:

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 48 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i,ele,f=0;
clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n); for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Enter search element "); scanf("%d",&ele);
for(i=0;i<n;i++)
{
if(a[i]==ele) printf("%d in a[%d] cell\n",ele,i,f=1);
}
if(f==0)printf("%d not found",ele);
getch();
}
```

Enter array size 1-100 7
Enter 7 elements 1 3 7 2 5 2 8
Enter search element 2
2 in a[3] cell
2 in a[5] cell

```

for(i=0; i<4; i++)
{
    if(a[i]==ele)
    p("%d in %d cell\n", ele, i+1, f=1);
}

```

7 3
 if(f==0) p(ele not found);

a			
<u>9</u>	<u>0</u>	<u>7</u>	<u>5</u>
0	1	2	3

<u>n</u>	<u>i</u>	<u>e e</u>	<u>f</u>
4	0	7	0
	1		1
	2+1		

20 0 ✓

Sorting:

Arranging data in ascending or descending order.

Selection sort in ascending order:

9	0	7	5
0	9	7	5
0	7	9	5
0	5	9	7
0	5	7	9

a

9	0	7	5
---	---	---	---

0 1 2 3

L R

i j

0 1, 2, 3

1 2, 3

2 3

```

for(i=0; i<=n-2;i++)
{
for( j=i+1; j<=n-1; j++)
{ L R
if(a[i]>a[j])
{
t=a[i];
a[i]=a[j];
a[j]=t;
}
}
}

```

a

9 0	0 7 5	7 9 7	7 5 9
0	1	2	3

L R

i j

0 1, 2, 3

1 2, 3

2 3

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 15 Col 36 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i,j,t;
clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n); for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<=n-2;i++)
{
for(j=i+1;j<=n-1;j++)
{
if(a[i]>a[j]){t=a[i]; a[i]=a[j]; a[j]=t;}}
printf("Sorted elements are ");
for(i=0;i<n;i++)printf("%4d",a[i]);
getch();
}
```

Enter array size 1-100 9
Enter 9 elements 8 -3 9 0 3 7 8 4 -2
Sorted elements are -3 -2 0 3 4 7 8 8 9_

Descending order:

The image shows a screenshot of a Turbo C++ IDE. The top window displays the source code for a program that sorts an array of integers. The code uses a bubble sort algorithm. The bottom window shows the program's execution, where the user enters an array size of 7 and the elements 2, 9, 0, 4, 8, 2, -4. The program then outputs the sorted elements: 9, 8, 4, 2, 2, 0, -4.

```
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 9 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i,j,t;
clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n); for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<=n-2;i++)
{
for(j=i+1;j<=n-1;j++)
{
if(a[i]<a[j]){t=a[i]; a[i]=a[j]; a[j]=t;}}
printf("Sorted elements are ");
for(i=0;i<n;i++)printf("%4d",a[i]);
getch();
}
```

Enter array size 1-100 7
Enter 7 elements 2 9 0 4 8 2 -4
Sorted elements are 9 8 4 2 2 0 -4

Print even elements in ascending order and odd elements in descending order.

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 16 Col 75 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i,j,t;
clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n); for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0;i<=n-2;i++)
{
for(j=i+1;j<=n-1;j++)
{
if(a[i]>a[j]){t=a[i]; a[i]=a[j]; a[j]=t;}}
printf("Sorted elements are ");
for(i=0;i<n;i++)if(a[i]%2==0){textcolor(LIGHTRED);cprintf("%4d",a[i]);}
for(n--;n>=0;n--)if(a[n]%2!=0){textcolor(LIGHTGREEN);cprintf("%4d",a[n]);}
getch();
}
}
```

Enter array size 1-100 9
Enter 9 elements 3 9 2 7 6 -1 4 -6 0
Sorted elements are -6 0 2 4 6 9 7 3 -1

```
for(i=0;i<6;i++)if(a[i]%2==0) p(a[i]);
```

```
for( ✓ n--; n>=0; n-- ) if(a[n]%2!=0) p(a[n]);
```

-4	0	7	10	11	15
0	1	2	3	4	5

7
6
5
4
3
2

-4 0 10 15 11 7

Home work:

1. arrange half elements in ascending and half descending
2. inserting a new element in specified position of array.

-4	0	<u>7</u>	10	11	
0	1	2	3	4	5

newele pos
9 3

3. deleting the specified element from array

-4	0	7	10	11	
0	1	2	3	4	5

ele

 7